STATISTICS AND PROBABILITY :

Introduction to statistics :

Statistics is the branch of applied mathematics and science concerned with collecting, organizing, analyzing, interpreting and presenting data.

Why is statistics used in machine learning ?

- → To understand data
- → To make better predictions
- → To handle uncertainty
- → To test if a model is better
- → To find relationships
- → To make real world decisions

There are two types of statistics :

① Descriptive Statistics :

Summarize and describe the data

② Inferential Statistics :

Make conclusions about a larger population

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① Descriptive statistics consists of:

1. Central Tendency (Mean, Median, Mode)
2. Dispersion of data (Range, Variance, Standard Deviation)
3. Frequency Distribution
4. Graphs / Charts
5. Quartiles & Percentiles
6. Outliers and Skewness

* Measures of Central Tendency:

- Central tendency is defined as the value around which data tends to cluster, the typical or central value

- Mean:

Mean is the average value calculated by dividing total of all values by number of values.

Mean: $\bar{X} = \frac{\Sigma x}{n}$ = Total of all values / Number of values

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E.g:

Marks of 20 students given;

$\Sigma x = 630 \qquad n = 20$

$\therefore \bar{x} = \frac{\Sigma x}{n} = \frac{630}{20} = 31.5$

In frequency table:

$\bar{x} = \frac{\Sigma fx}{n}$

E.g:

Marks(x)	Frequency	fx
50	2	100
60	3	180
70	5	350
80	4	320
90	3	270
100	3	300
	$n = 20$	$\Sigma fx = 1520$

$\bar{x} = \frac{\Sigma fx}{n} = \frac{1520}{20} = 76$

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Why is mean used in data science?

→ Data summarization : Mean gives a single value to summarize large data

→ Missing value Filling : Mean is often used to impute missing numerical values

→ Standardization (Feature scaling) : Mean is used to center the data (subtract mean) in standardization (z-score normalization).

~~Flaws~~ ~~Weakness~~ of mean :

1. Highly sensitive to outliers
2. Misleading in skewed data

→ Group Comparison : Compares the average of different groups

→ Outlier detection : values far from the mean can be flagged as potential outliers.

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- Median:

→ Median is ~~the~~ defined as the middle value of the sorted data.

When the number of observations is odd:

Median = Value at position $\frac{n+1}{2}$

When the number of observations is Even:

median is the average of two middle values

$$\text{Median} = \frac{\text{value at } n/2 + \text{value at } (n/2 + 1)}{2}$$

Eg:

10, 20, 30, 40, 50, 60

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E.g:

Q// 10, 20, 30, 40, 50, 60

→ Median = $\frac{30+40}{2} = 70/2 = 35$

- Mean Vs Median:

E.g:

Q// Given salaries: million

50, 75, 120, 135, 260, 24,000 (in dollars)

$$\text{Mean } (\bar{x}) = \frac{50+75+120+135+260+24000}{6}$$

$$= \frac{24640}{6}$$

$$= \$4106.7$$

∴ Mean $(\bar{x})$ = \$4106.7 million

= \$4.1 billion

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median = $\frac{120 + 135}{2}$

= $127.5 million

∴ comparing both answers, we can observe that one extreme value can ruin the value of mean whereas the median gives a sensible response.

- Cumulative Frequency (cf):

→ Cumulative frequency is the running total of frequencies. It tells us how many observations are less than or equal to a particular value.

→ The cumulative frequency shows the total number of observations upto and including that value.

→ It helps us to find the position of that ~~value~~ particular observation

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- Mode:

The value with the highest frequency

E.g: (8)

Suppose a data occurs 35 times in a table. No other data is repeated in the table above 35 times. The mode is 8.

Popularity: It is the most common choice of the users

(*) Measures of Dispersion:

1. Range
2. Variance
3. Standard deviation
4. InterQuartile Range (IQR)

- Range:

Range shows the spreadness of data

Range = Maximum - Minimum

- Larger range means more spread,
Smaller range means less spread

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E.g:

Max = 30

Min = 10

∴ Range = 30 - 10

= 20

- Variance:

Variance shows how spread the values are around the mean.

Formula: (σ^2)

$$\text{variance} = \frac{\Sigma fx^2}{n} - \left(\frac{\Sigma fx}{n}\right)^2$$

OR (σ^2)

$$\text{variance} = \frac{\Sigma (a - \bar{x})^2}{n}$$

- variance measures spread around the mean
- Smaller variance → value stays close to mean → More consistency
- Larger variance → value spreads far from the mean → More variability

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Eg:

Data values (x):

48, 49, 50, 51, 52

$n = 5$

$$\text{mean } (\bar{x}) = \frac{48+49+50+51+52}{5}$$

$$= \frac{250}{5} = 50$$

x	x - mean	$(x-\text{mean})^2$
48	$48-50 = -2$	4
49	$49-50 = -1$	1
50	$50-50 = 0$	0
51	$51-50 = 1$	1
52	$52-50 = 2$	4
		$\Sigma(x-\text{mean})^2 = 10$

$$\therefore \text{Variance } (\sigma^2) = \frac{\Sigma(x-\text{mean})^2}{n}$$

$$= \frac{10}{5} = 2$$

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• Standard Deviation :

It tells us how much the data values typically differ from the mean.

→ It measures the average distance of each value from the mean

→ Small Standard Deviation → values are close to mean

→ Large Standard Deviation → values are spread out

Formula :

$$\text{Standard Deviation } (\sigma) = \sqrt{\text{variance}}$$

$$= \sqrt{\frac{\Sigma (x - \text{mean})^2}{n}}$$

$$= \sqrt{\frac{\Sigma fx^2}{n} - \left(\frac{\Sigma fx}{n}\right)^2}$$

Why Standard Deviation?

→ variance gives the average of the squared differences from the mean, but its unit is squared which is hard to interpret.

→ Standard deviation is easier to understand and interpret in real-world situations as it is the squared-root of variance, so it is the same unit as the original data

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- Interquartile Range (IQR):

- Percentile:

Percentile tells us the position of a value in sorted data.

→ The pth percentile means p% of the data is at or below that value

→ Higher percentile means higher position in the dataset.

E.g.:

Given sorted marks:

40, 45, 50, 55, 60, 65, 70, 75, 80, 90

70th percentile = 70

70% of values are at or below 70.

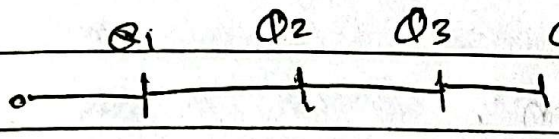
- 25th percentile = lower part
- 50th percentile = median
- 75th percentile = upper part

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Quartiles:

Sorted data are divided into four equal parts called quartiles. with n observation.

- Q_1 = 25th percentile $\left(\frac{n+1}{4}\right)$
- Q_2 = 50th percentile $\left(\frac{n+1}{2}\right)$
- Q_3 = 75th percentile $\left\{3\left(\frac{n+1}{2}\right)\right\}$
- Q_4 = 100th percentile (The maximum value)



E.g:

Given Sorted data,

2, 4, 5, 8, 10, 12, 15 $\quad n = 7$

$Q_1 = \frac{7+1}{4} = 8/4 = 2$

$Q_2 = \frac{7+1}{2} = 8/2 = 4$

$Q_3 = 3\left(\frac{7+1}{2}\right) = 24/2 = 12$

$Q_4 = 15$ (max val.)

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∴ Inter-quartile Range (IQR)

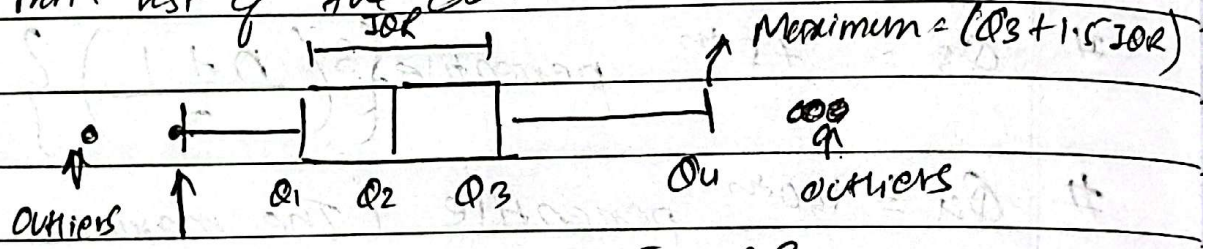
The spread of the middle 50% of the data.

Formula:

$$IQR = Q_3 - Q_1$$

• Outliers:

An outlier is a value unusually far from rest of the data



Minimum $= Q_1 - 1.5 \times IQR$

• Outlier Detection:

An outlier is a value unusually far from rest of the data.

$IQR = Q_3 - Q_1$

Lower Bound $= Q_1 - 1.5\ IQR$

Upper Bound $= Q_3 + 1.5\ IQR$

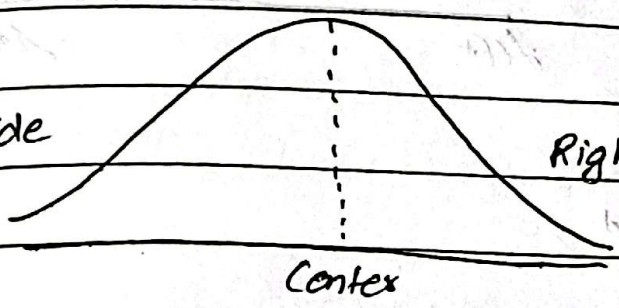
Any value below the lower bound or above the upper bound is potential outlier.

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Symmetric Distribution:

A symmetric distribution is a distribution where the data is spread almost equally on both sides of the centre.

→ The left and right side look like mirror images.



For perfect Symmetric distribution:

Mean = Median = Mode

In a symmetric distribution, the values on the left and right sides of the centre are distributed similarly.

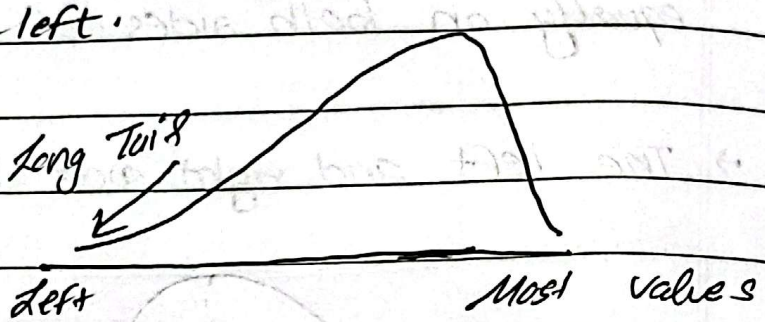
Skewness:

It describes the asymmetry of distribution.

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- **Left - skewed Distribution:**

A left skewed distribution is a distribution where most of the data is concentrated on the right side, while a few very small values create a long tail on the left.



$\bar{x} < Md$

- **Right - skewed Distribution:**

Most values are low, a few very large value form a long tail on the right.

$\bar{x} > Md$

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Covariance

Covariance shows whether two variables move together in the same direction or in opposite directions.

formula:

$$cov(x, y) = \frac{\Sigma (x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

Interpretation:

→ cov > 0 : Positive Covariance
→ cov < 0 : Negative Covariance
→ cov ≈ 0 : No clear linear relationship

Covariance tells the direction of the relationship between two variables
Positive covariance means both variables tend to increase together
Negative covariance means one increases while the other decreases
covariance is not standardized, so its size alone does not directly tell the strength like correlation does.

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Correlation :

Correlation measures the strength and direction of the linear relationship between two quantitative variables. It tells us how one variable changes with respect to another.

Formula :

$$r = \frac{\Sigma (X - \bar{X})(Y - \bar{Y})}{\sqrt{\Sigma (X - \bar{X})^2} \sqrt{\Sigma (Y - \bar{Y})^2}}$$

where :

- X : value of first variable
- Y : value of second variable
- $\bar{X}$: Mean of X
- $\bar{Y}$: Mean of Y
- r : Pearson correlation coefficient

Keypoints :

- -> Value ranges from -1 to +1
- -> +1 : Perfect positive linear relationship
- -> -1 : Perfect negative linear relationship
- -> 0 : No linear relationship
- -> It is unitless (no unit)
- -> It measures only linear relationship, not causation.

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→ It is based on covariance and standard deviation.

• Example:

$$\Sigma(x-\bar{x})(y-\bar{y}) = 6$$
$$\Sigma(x-\bar{x})^2 = 10$$
$$\Sigma(y-\bar{y})^2 = 6$$

So Now,

$$r = \frac{\Sigma(x-\bar{x})(y-\bar{y})}{\sqrt{\Sigma(x-\bar{x})^2}\sqrt{\Sigma(y-\bar{y})^2}}$$

$$= \frac{6}{\sqrt{10 \times 6}}$$

$$= \frac{6}{\sqrt{60}} = 6/7.746$$

$\therefore r \approx 0.775$

∴ Since r is positive and fairly close to 1, study Hours and Marks have a fairly strong positive linear relationship.

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o Z-score and standardization:

→ A z-score tells how many standard deviations a value is above or below the mean.

→ Standardization converts a value into a z-score.

Formula:

$$z = \frac{x - \mu}{\sigma}$$

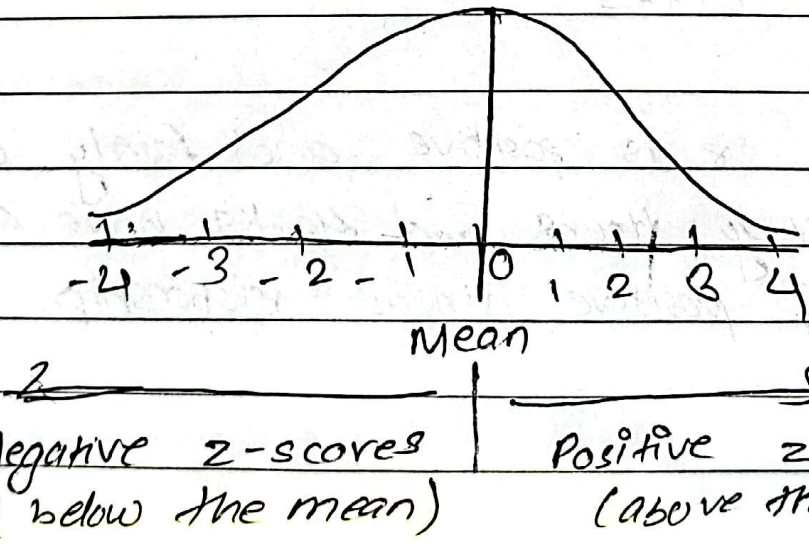
where,

x = actual value

μ = mean

σ = standard deviation

z = z-score



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Interpretation of z-scores:

- $z > 0$: value is above the mean
- $z < 0$: value is below the mean
- $z = 0$: value is exactly at the mean
- larger $|z|$: farther from the mean
- smaller $|z|$: closer to the mean

Standardization:

Converts each value to a z-score giving mean 0 and standard deviation 1.

- Makes different variables comparable
- Removes the original units
- Helps detect unusual values
- Widely used in machine learning and statistics

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- Normalization:

Rescales values to a fixed range usually 0 to 1.

Formula:

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}}$$

Where,

x = original value

x_{min} = minimum value

x_{max} = maximum value

x' = normalized value

E.g:

Age = 25 (min = 18, max = 35)

$$x' = \frac{25 - 18}{35 - 18} = \frac{7}{17} \approx 0.41$$

- Normalization rescales values to the range 0 to 1.

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- when to use which :

Normalization :

when you need a fixed range from 0 to 1. Good for image pixels and some neural network inputs. Often useful for KNN and K-means.

Standardization :

When features have different unit or scales. It centers data around mean 0 and scales data to standard deviation 1. Common for Linear Regressions, Logistic regression, SVM and PCA.

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Normal Distribution :

A bell-shaped, symmetric, continuous probability distribution where most values are near the means and extremes are rare. It is written $X \sim N(\mu, \sigma^2)$

Properties :

1) Bell-shaped, smooth, symmetric arand the mean

2) Mean : Median = Mode

3) Total area under the curve = 1 ;
50 % of the data lies on each side of the mean

4) Fewer values appear as you move away from the mean

5) Completely defined by two parameters: μ controls the center and σ controls the spread.

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Formula:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \cdot \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

$$X \sim N(\mu, \sigma^2)$$

where,

x = a data value

μ = mean (centre of the distribution)

σ = standard deviation

σ^2 = variance

π = 3.14

e = 2.72

- Empirical rule (68-95-99.7):

→ About 68% of value lies within $\mu \pm 1\sigma$

→ About 95% of value lies within $\mu \pm 2\sigma$

→ About 99.7% of values lies within $\mu \pm 3\sigma$

→ Values beyond $\mu \pm 3\sigma$ are very rare and maybe possible outliers.

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- Central Limit Theorem:

If you take many random samples and compute each sample mean, the distribution of these sample means is approximately normal, even if the original data is skewed or uniform.

- Probability Basic:

Probability measures how likely an event is to happen.

Formula: $P(A) = \frac{\text{Favourable outcomes}}{\text{Total outcomes}}$

Range: $0 \leq P(A) \leq 1$

→ 0 = impossible (rolling a 10 on a normal dice)

→ 0.5 = equally likely (heads on fair coin)

→ 1 = certain (rolling a number from 1 to 6)

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Example :

-> P (Even on a dice) : $3/6 = 0.5$

-> P (Ace from 52 cards) : $4/52$

$= 1/13$

$= 0.0769$

Rules of Probability :

- Range rule : $0 \leq P(A) \leq 1$
- complement rule : $P(A') = 1 - P(A)$
- Addition rule (A or B) : $P(A) + P(B)$

$= P(A) + P(A \text{ and } B) - P(A \text{ or } B)$

Example :

$P(\text{roll 1 or 6}) = 1/6 + 1/6$

$= 2/6 = 1/3$

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* Multiplication Rule:

$$P(A \cap B) = P(A) \cdot P(B|A)$$

-> If A and B are independent:

$$P(A \cap B) = P(A) \cdot P(B)$$

- Independent variable Example:

$$P(\text{Two sixes in a row}) = \frac{1}{6} \times \frac{1}{6}$$

$$= \frac{1}{36}$$

- Dependent variable:

$$P(\text{Two aces in a row without replacement}) = \frac{4}{52} \times \frac{3}{51}$$

$$= \frac{1}{221}$$

(After one Ace is drawn, {52-1=51}, only 3 Aces remain in 51 cards).

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* Conditional Probability:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

* Total Probability:

$$P(A) + P(A') = 1$$

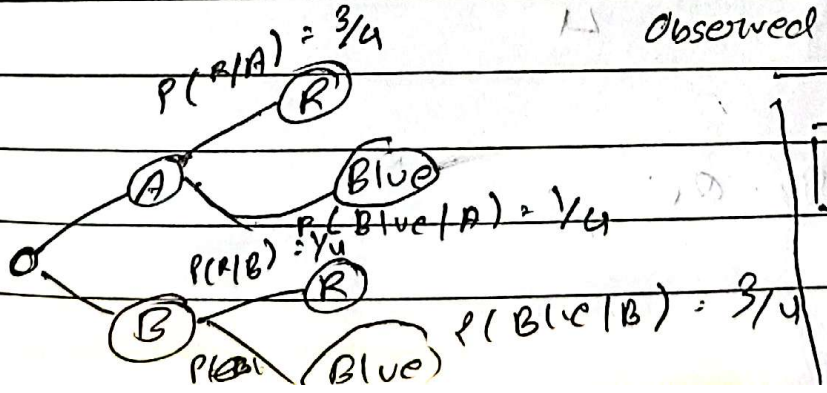
* Bayes's Theorem:

Bayes's theorem updates the probability of a hypothesis after seeing new evidence.

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

E.g.:

$P(A) = 1/2$ $\quad P(B) = 1/2$



Observed evidence : R

A

R	R
R	B

B

B	B
B	R

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$$P(A|R) = \frac{P(R|A) \times P(A)}{P(R)} = \frac{3/4 \times 1/2}{1/2} = 3/4$$

$$P(R) = P(R|A)\ P(A) + P(R|B)\ P(B)$$

$$= 3/4 \times 1/2 + 1/4 \times 1/2 = 1/2$$

(*) Quick Formula Sheet:

- Mean = $\Sigma x / n$
- Median Position = $\frac{(n+1)}{2}$
- Range = Max − Min
- variance = $\frac{\Sigma (x - \text{mean})^2}{N}$
- S.D. = $\sqrt{\text{variance}}$
- Quartile Position = ~~100~~ ~~Q1~~

$$Q_1 = \frac{n+1}{4}, \quad Q_2 = \frac{n+1}{2}, \quad Q_3 = \frac{3(n+1)}{4}$$

- IQR = $Q_3 - Q_1$

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- $Cov(X, Y) = \frac{\Sigma (x - \bar{x})(y - \bar{y})}{n-1}$
- $r = \frac{Cov(X, Y)}{\sigma(x) \cdot \sigma(y)}$
- $z = \frac{x - \mu}{\sigma}$
- min-max = $(x - min)(max - min)$
- Empirical rule = 68% / 95% / 99.7%
- P(A) = Favourable / Total
- P(A') = 1 - P(A)
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- $P(A \cap B) = P(A) \cdot P(B|A)$
- $P(A|B) = P(A \cap B) / P(B)$
- Bayes $\Rightarrow P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$

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